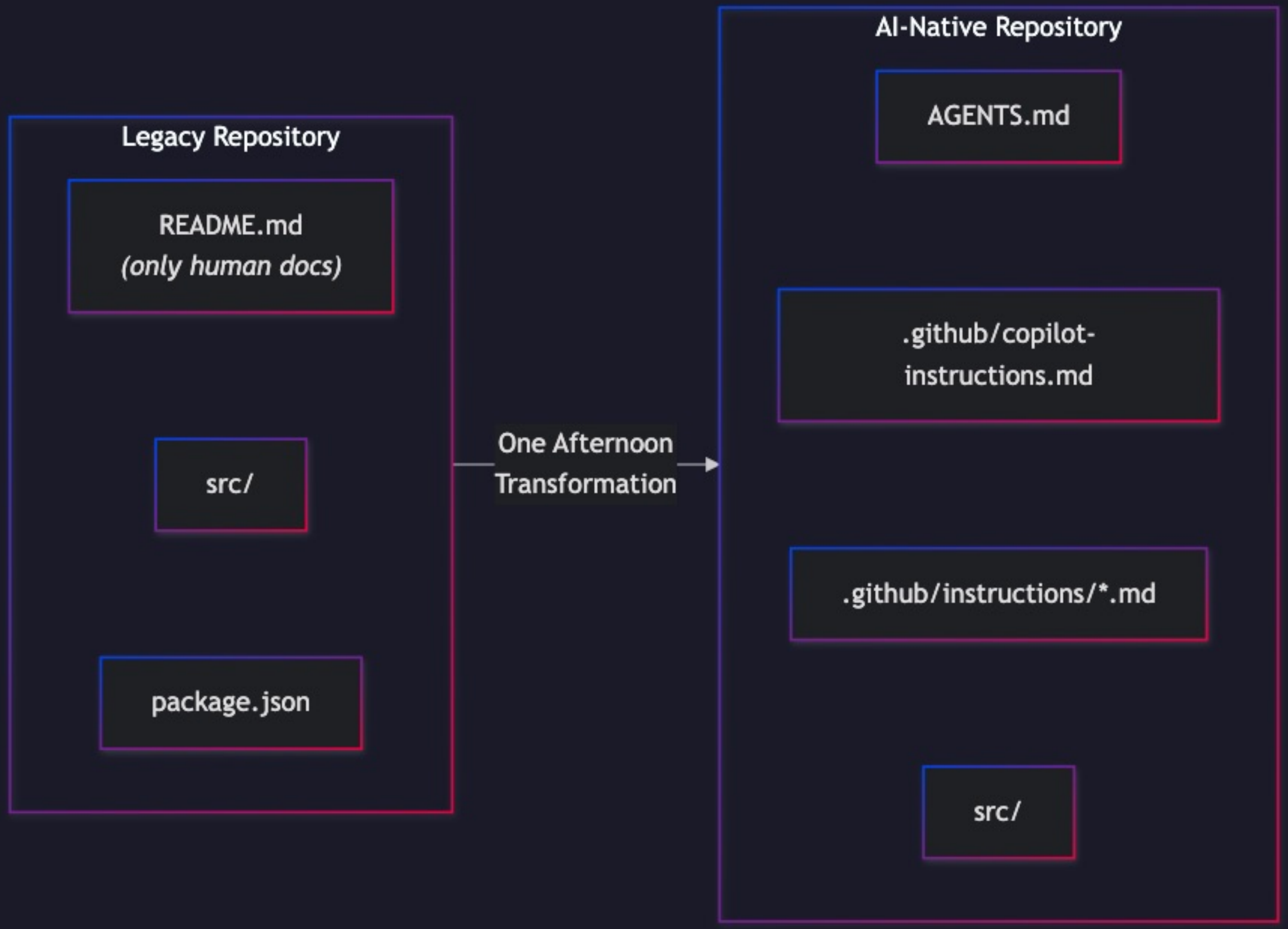
OPENING

"Your Repo Is Fighting Your AI Agents"

70% of AI coding tasks fail — not because models are bad, but because **context** is wrong. A few files fix it.

53%	100%	1 afternoon
Baseline task success	AI-native repo success	Time to transform

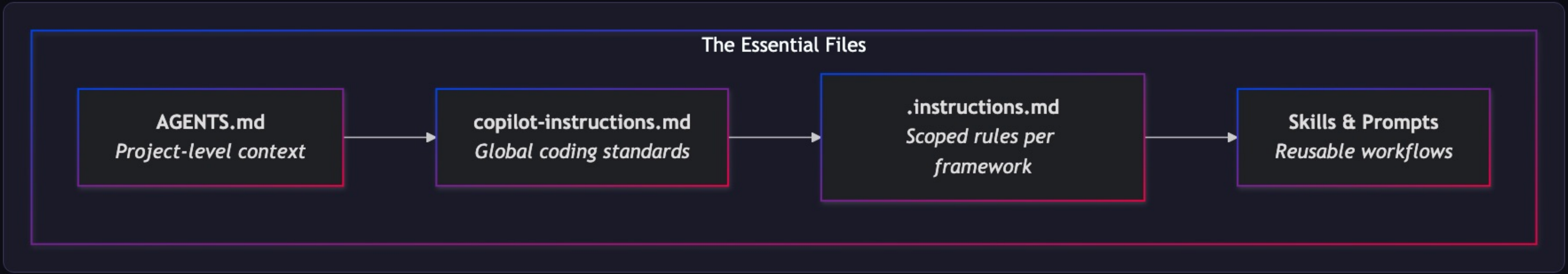


Vercel Blog, Jan 2026

PRACTICAL

The Essential Files — Your AI Agent Toolkit

Four files that every major AI coding agent reads. Add these and your agent stops guessing.



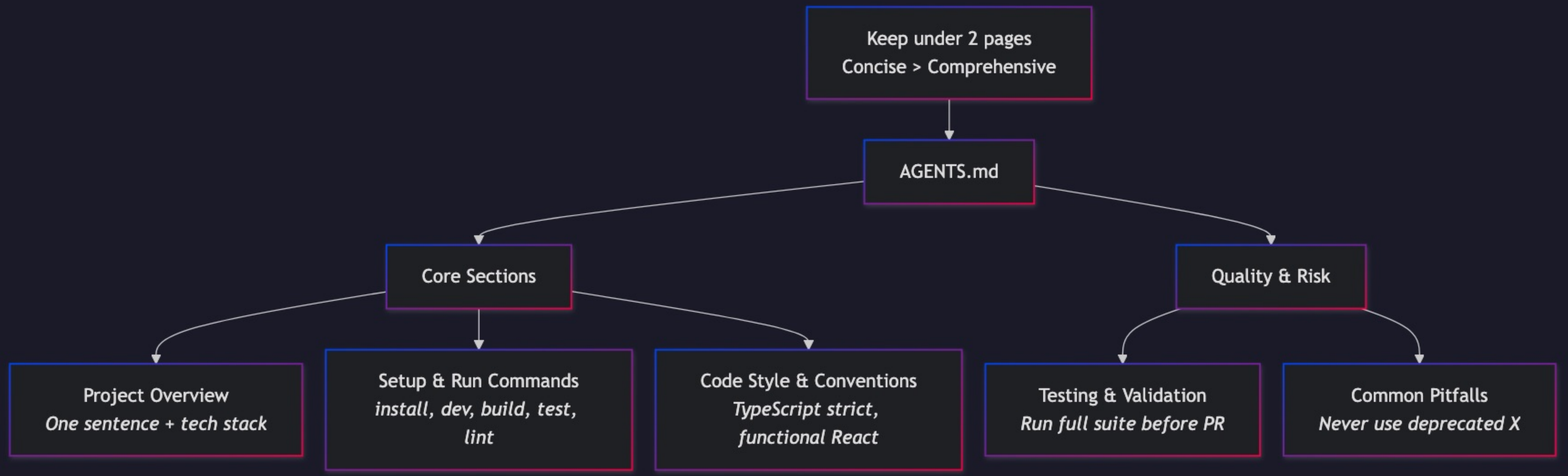
AGENTS.md Spec

GitHub Copilot Docs

PRACTICAL

What Goes in AGENTS.md

Five sections. Keep under 2 pages. Concise > Comprehensive. 8KB beat 40KB in Vercel's eval.



CRITICAL RULE

An 8KB compressed index beat 40KB of full docs. Encode your **tribal knowledge** — the non-obvious conventions, the things that burn people.

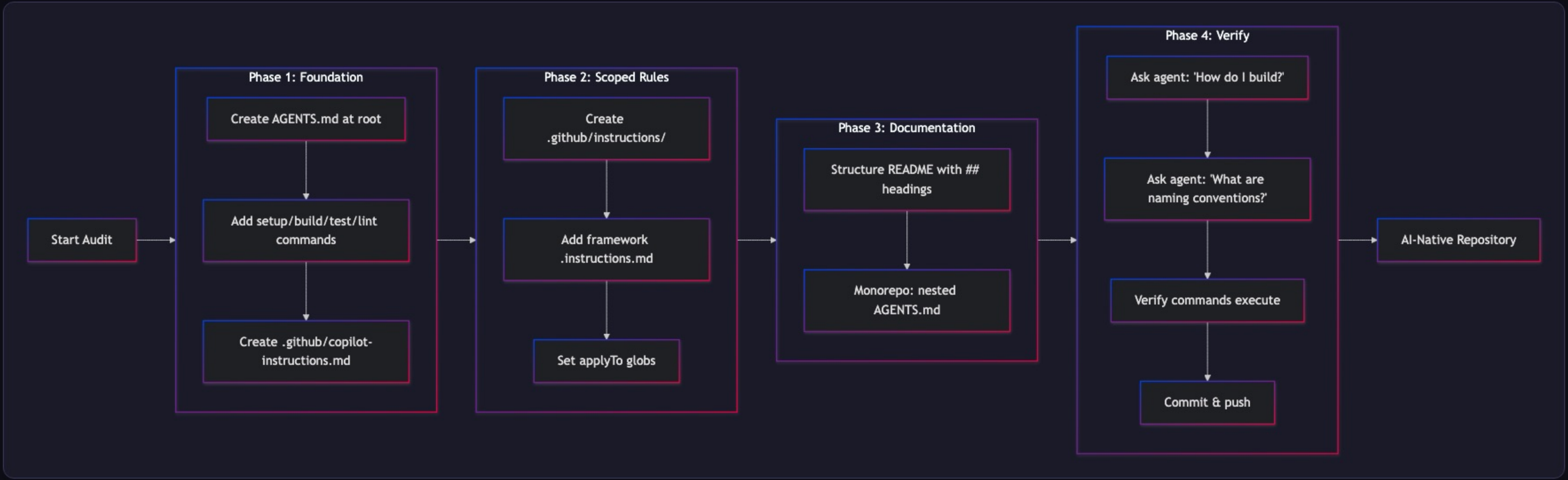
AGENTS.md Spec

GitHub Blog

PRACTICAL

The 15-Minute Audit

Four phases. Foundation → Scoped Rules → Documentation → Verify. You can do this during lunch.



PRACTICAL

Example Repo in Action

"acme-webapp" — a typical monorepo with all the essential files integrated.

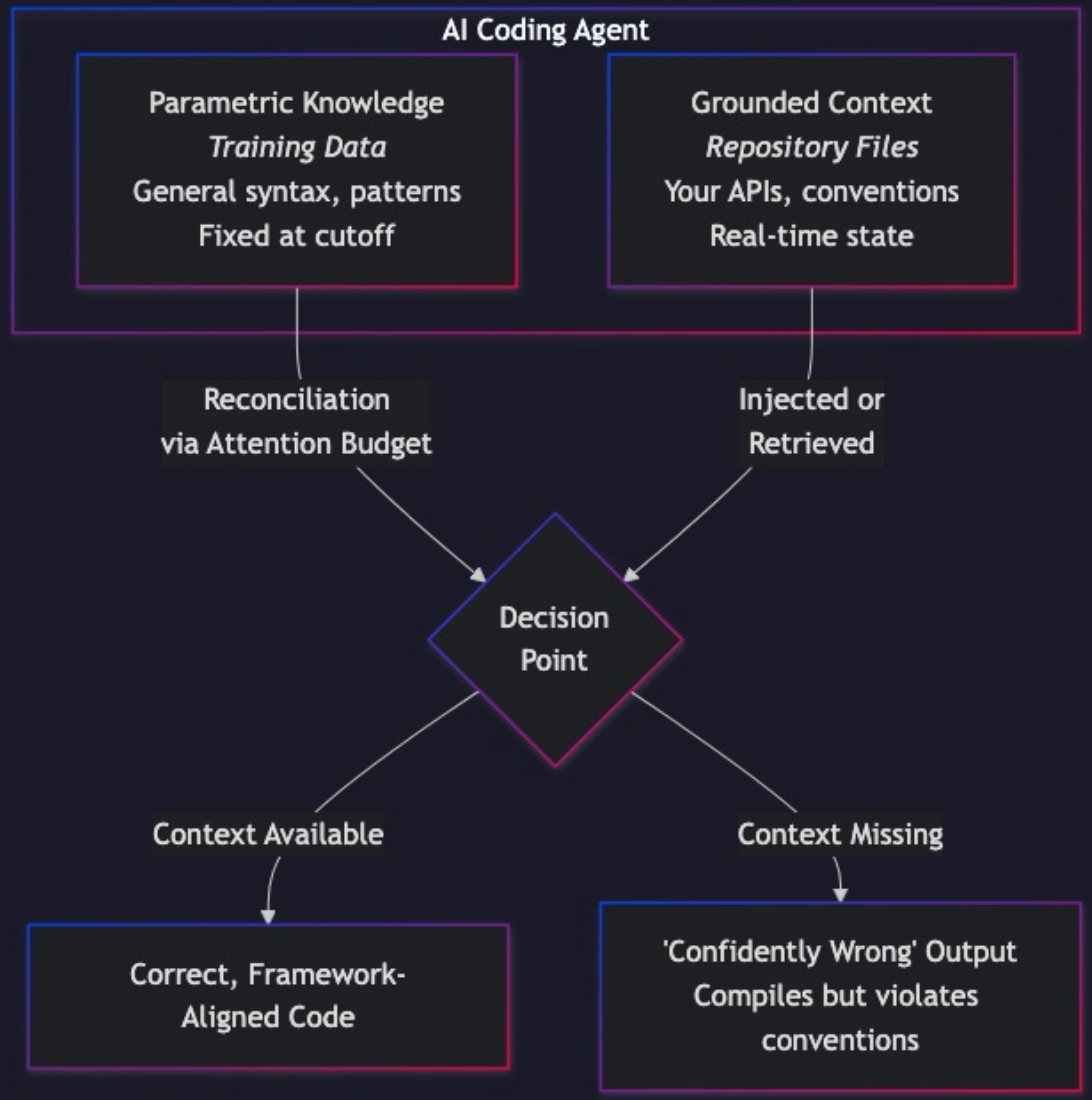
WHAT'S INSIDE

AGENTS.md — project overview & commands. **copilot-instructions.md** — global coding standards. **Scoped rules** — per-framework globs. **Prompts** — team-shared templates. **Skills** — reusable agent workflows.

THE WHY

Parametric vs. Grounded Knowledge

When grounded context is missing, parametric knowledge fills the gap — **confidently wrong**.



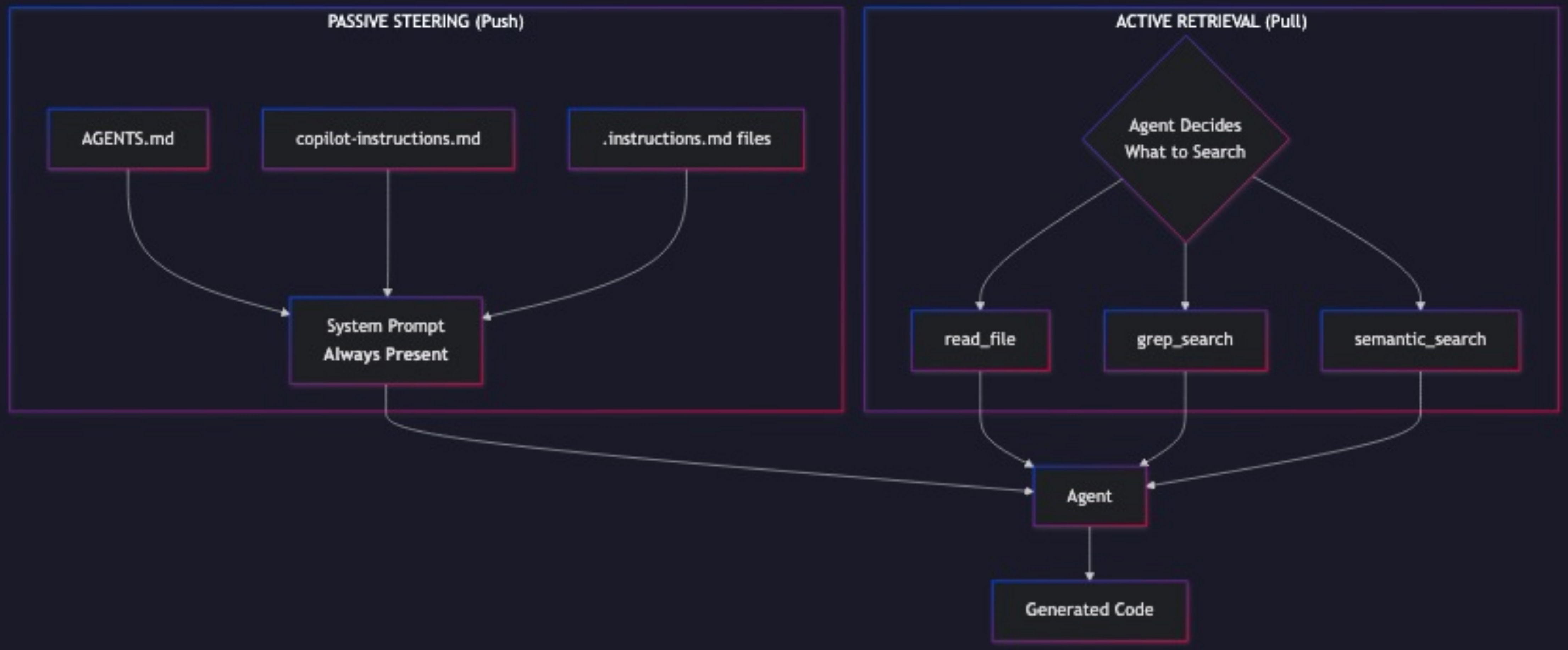
KEY INSIGHT

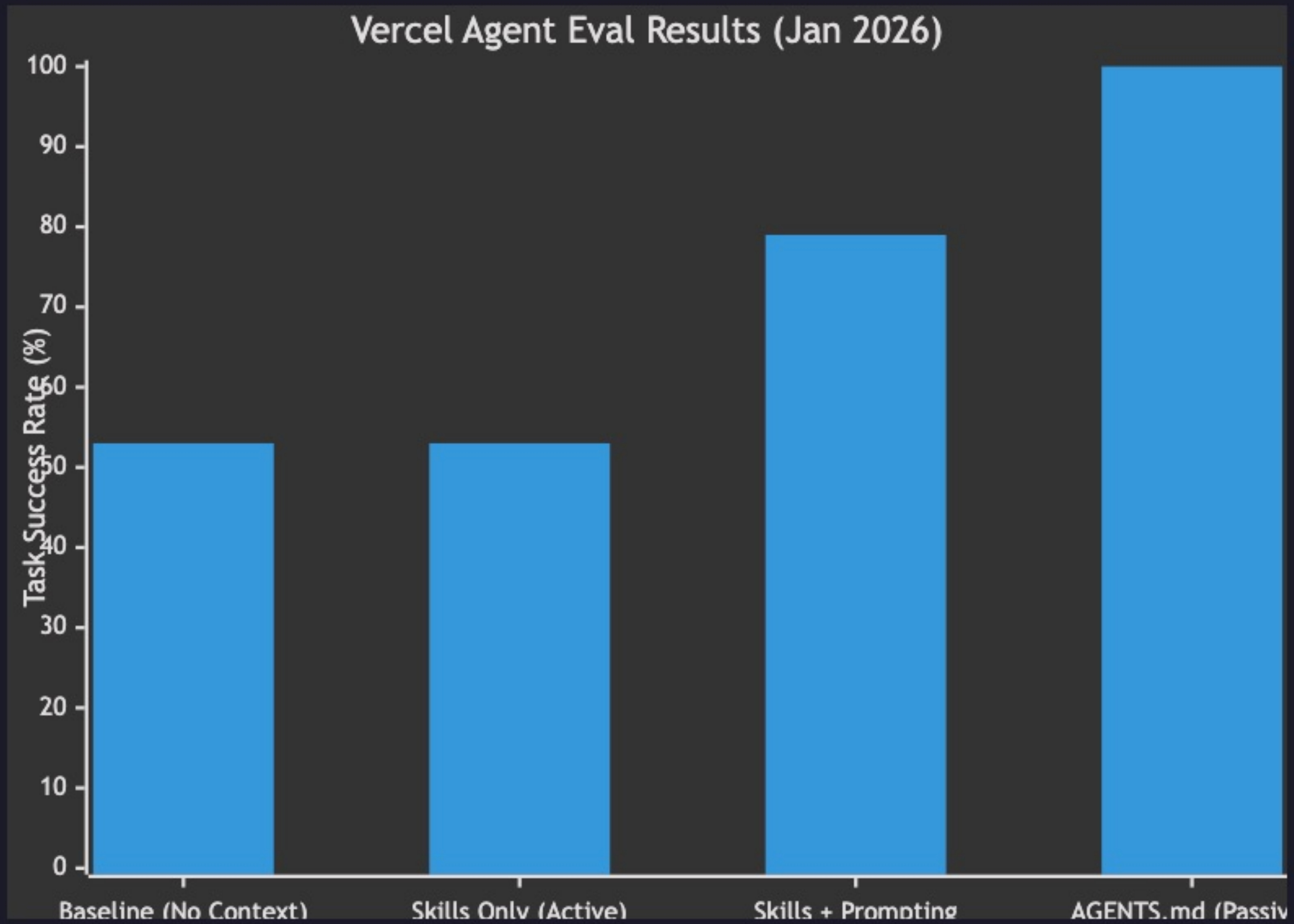
Your private APIs, naming conventions, monorepo structure — none of that is in the training data. The files we just set up provide that grounded context.

THE WHY

Why Passive Wins

Passive (push) vs. Active (pull). When agents decide whether to search, they skip docs 44% of the time.





CRITICAL RESULT

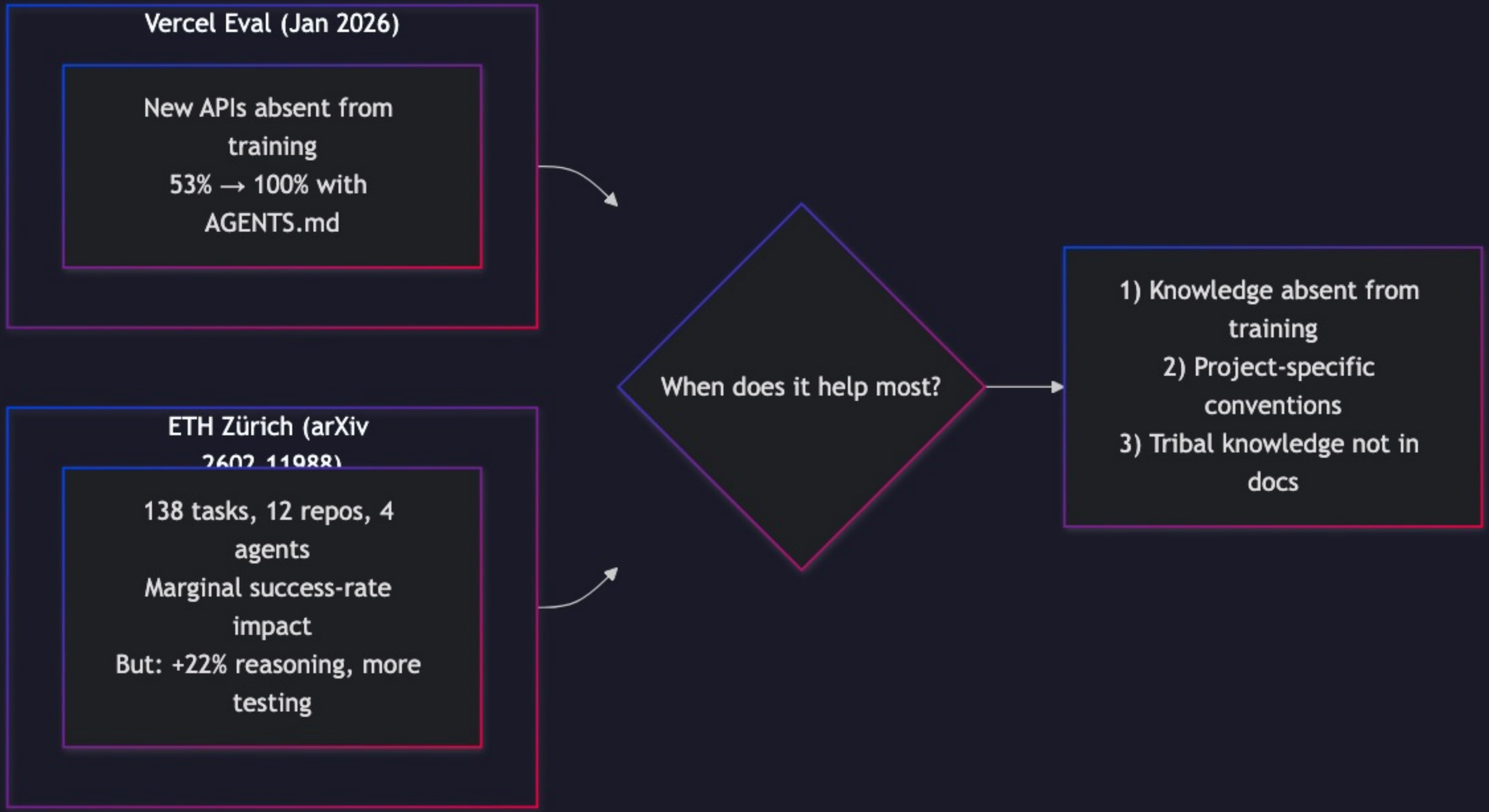
A compressed **8KB AGENTS.md** achieved **100% success rate —** beating 40KB of full docs via Skills (53%).

Vercel Blog, Jan 2026

THE WHY

Research **Snapshot**

Vercel shows dramatic gains. ETH Zürich finds marginal effects. The reconciliation is simple.



PRACTICAL TAKEAWAY

Don't auto-generate AGENTS.md with an LLM — those are redundant. **Write it yourself.** Encode your *tribal knowledge*: the stuff not in any documentation.

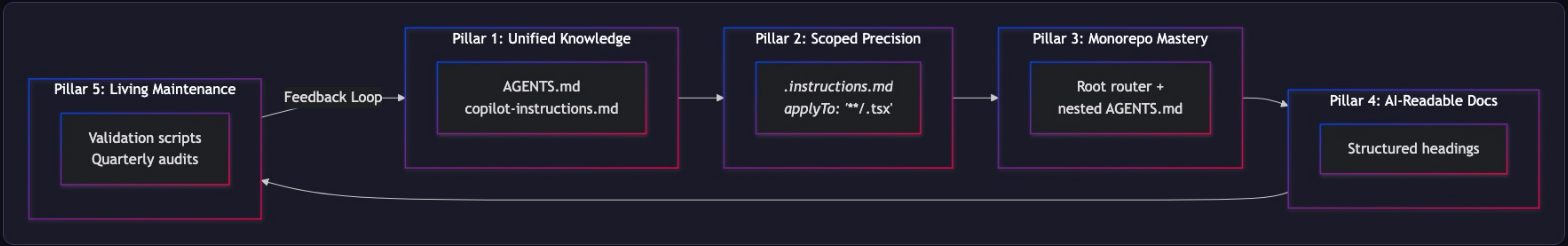
Vercel Blog

arXiv 2602.11988

ARCHITECTURE

The Five Pillars of AI-Native Repos

Unified Knowledge → Scoped Precision → Monorepo Mastery → AI-Readable Docs → Living Maintenance



WHERE YOU ARE

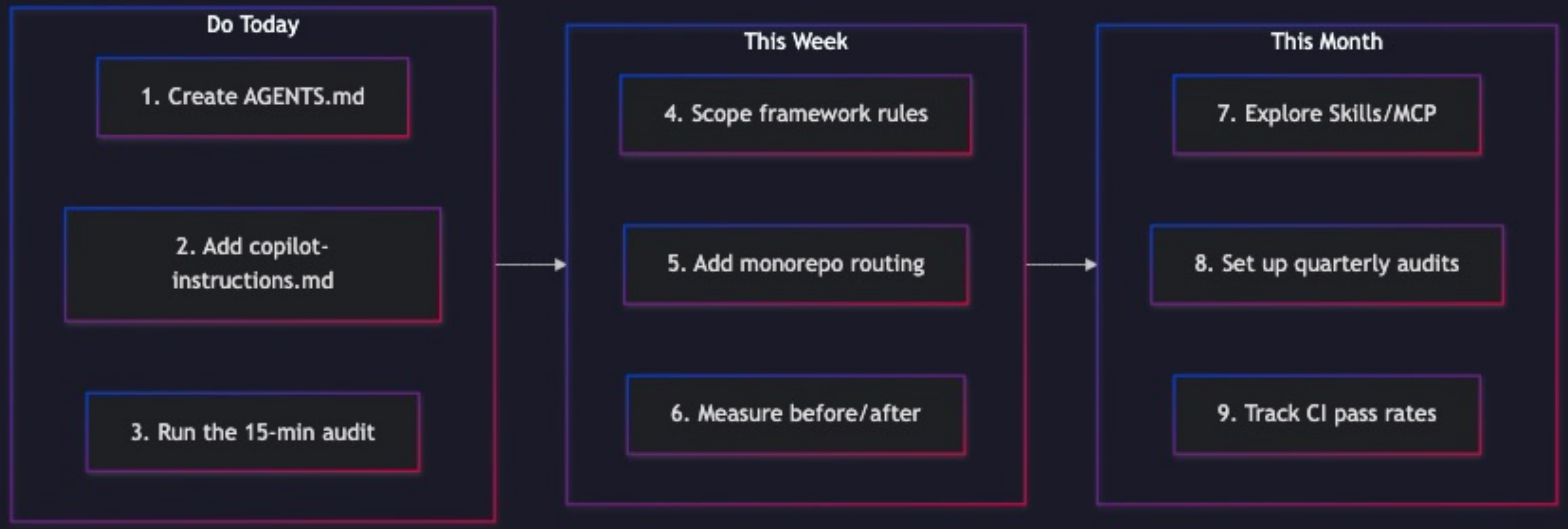
You already know how to do Pillars 1–3 from the audit slide. Pillars 4 & 5 are your this-week and this-month work.

AGENTS.md Spec

CALL TO ACTION

Maturity Model & Call to Action

Where is your repo today? Most teams are at Level 1. Getting to Level 3 takes an afternoon.



TODAY — 15 MIN
Create AGENTS.md, copilot-instructions.md, run the audit

THIS WEEK
Scope framework rules, add monorepo routing, measure before/after

THIS MONTH
Explore Skills/MCP, quarterly audits, track CI pass rates

REFERENCES

Key Citations & Sources

Primary research, specifications, and documentation referenced throughout this talk.

[1] Vercel Blog — *AGENTS.md outperforms skills in our agent evals* (Jan 27, 2026)
`vercel.com/blog/agents-md-outperforms-skills-in-our-agent-evals`

[2] arXiv 2602.11988 — *Evaluating AGENTS.md: Are Repository-Level Context Files Helpful for Coding Agents?* (ETH Zürich)
`arxiv.org/abs/2602.11988`

[3] AGENTS.md Specification — Official specification for repository-level agent instruction files
`agents.md`

[4] GitHub Docs — *Adding custom instructions for GitHub Copilot*
`docs.github.com/.../adding-custom-instructions-for-github-copilot`

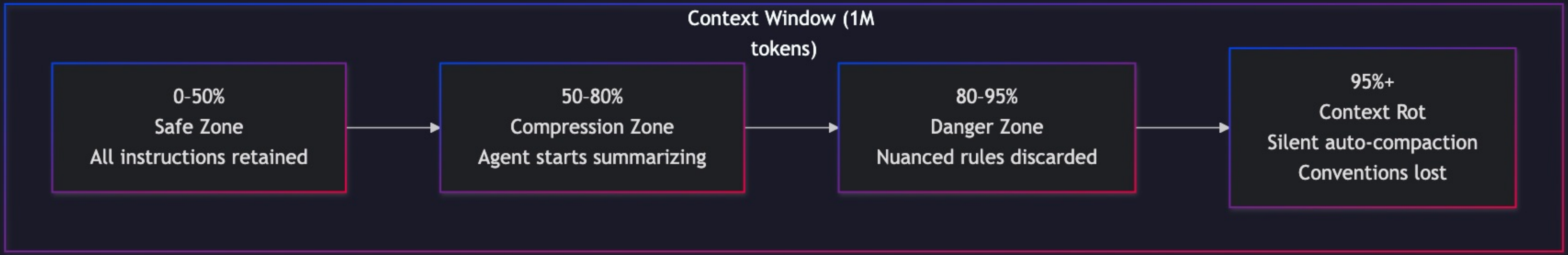
[5] Anthropic — *The Complete Guide to Building Skills for Claude*
`platform.claude.com/.../agent-skills/best-practices`

[6] GitHub Blog — *How to write a great agents.md — Lessons from over 2,500 repositories*
`github.blog/.../how-to-write-a-great-agents-md...`

APPENDIX A

Context Rot — The Silent Killer

Even million-token windows have a ticking time bomb. Agents silently auto-compact — your conventions are the first things discarded.



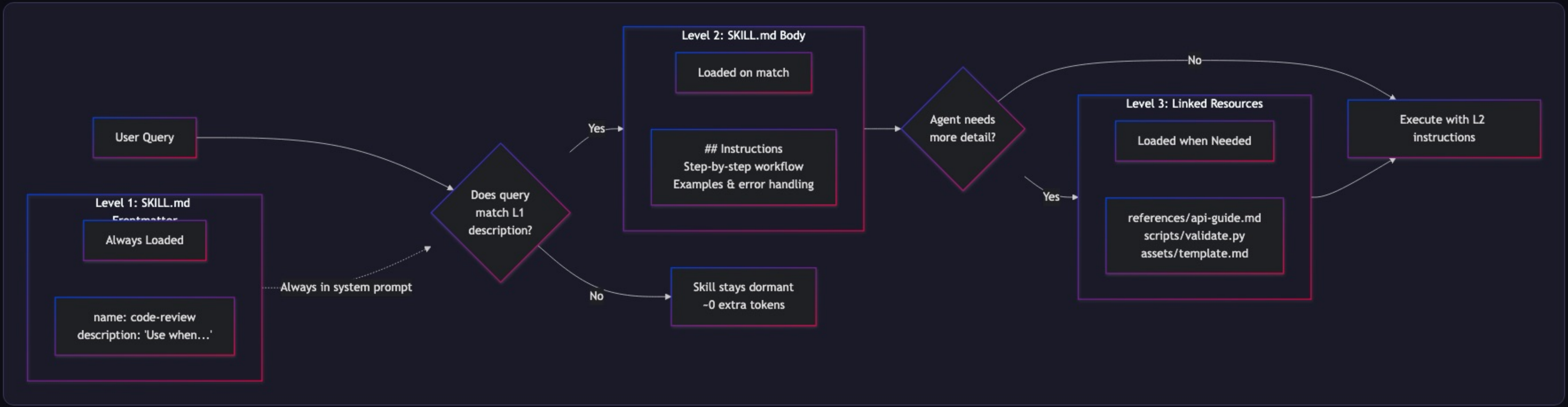
WHY THIS MATTERS

The code compiles. Tests might pass. But it silently violates your architecture. This is why we need **structured context**.

APPENDIX B

Anthropic's Progressive Disclosure

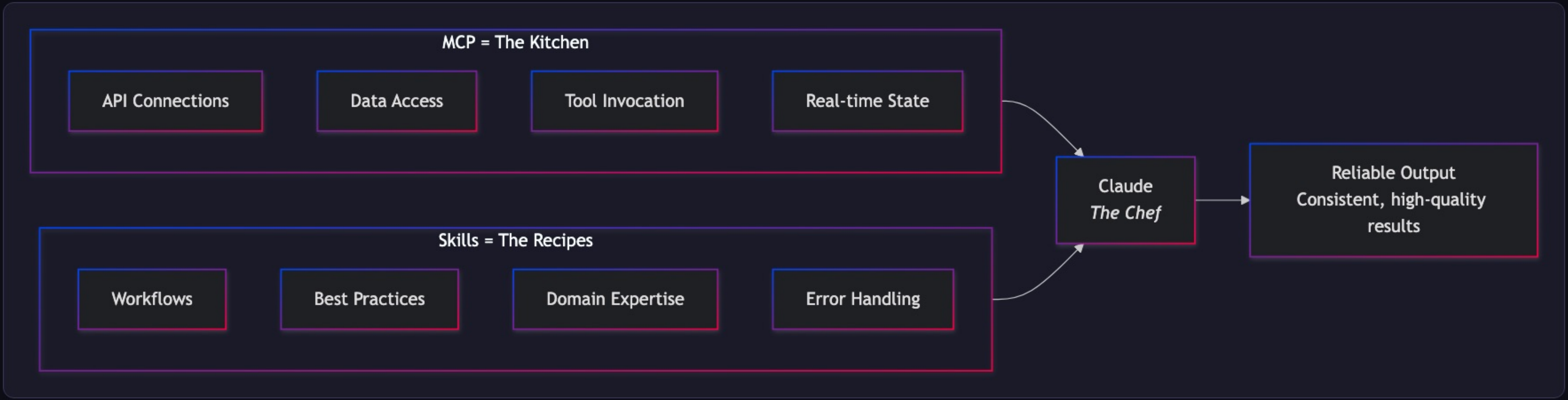
Three-level loading: YAML frontmatter → SKILL.md body → Linked resources. Hundreds of skills, minimal token overhead.



APPENDIX C

The MCP + Skills Architecture

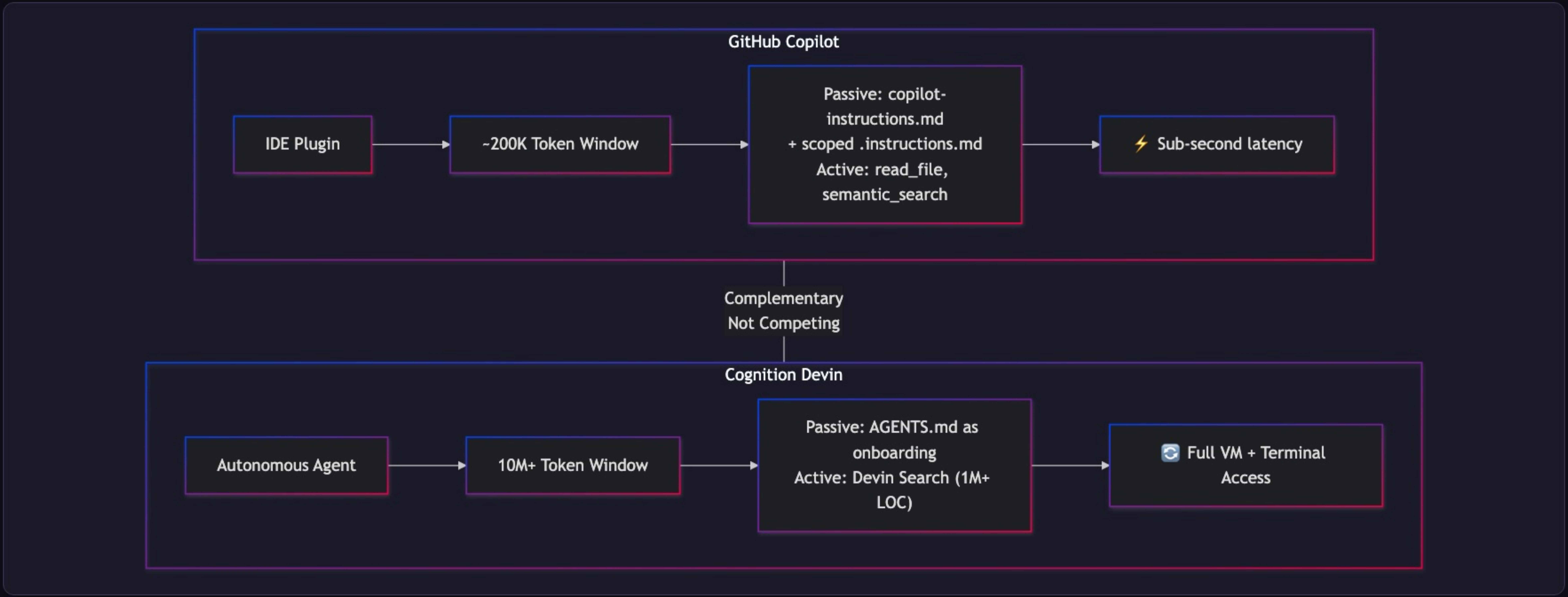
MCP provides capabilities (the kitchen). Skills provide workflows (the recipes). Together: reliable, scalable AI development.



APPENDIX D

Copilot vs. Devin — Two Architectures

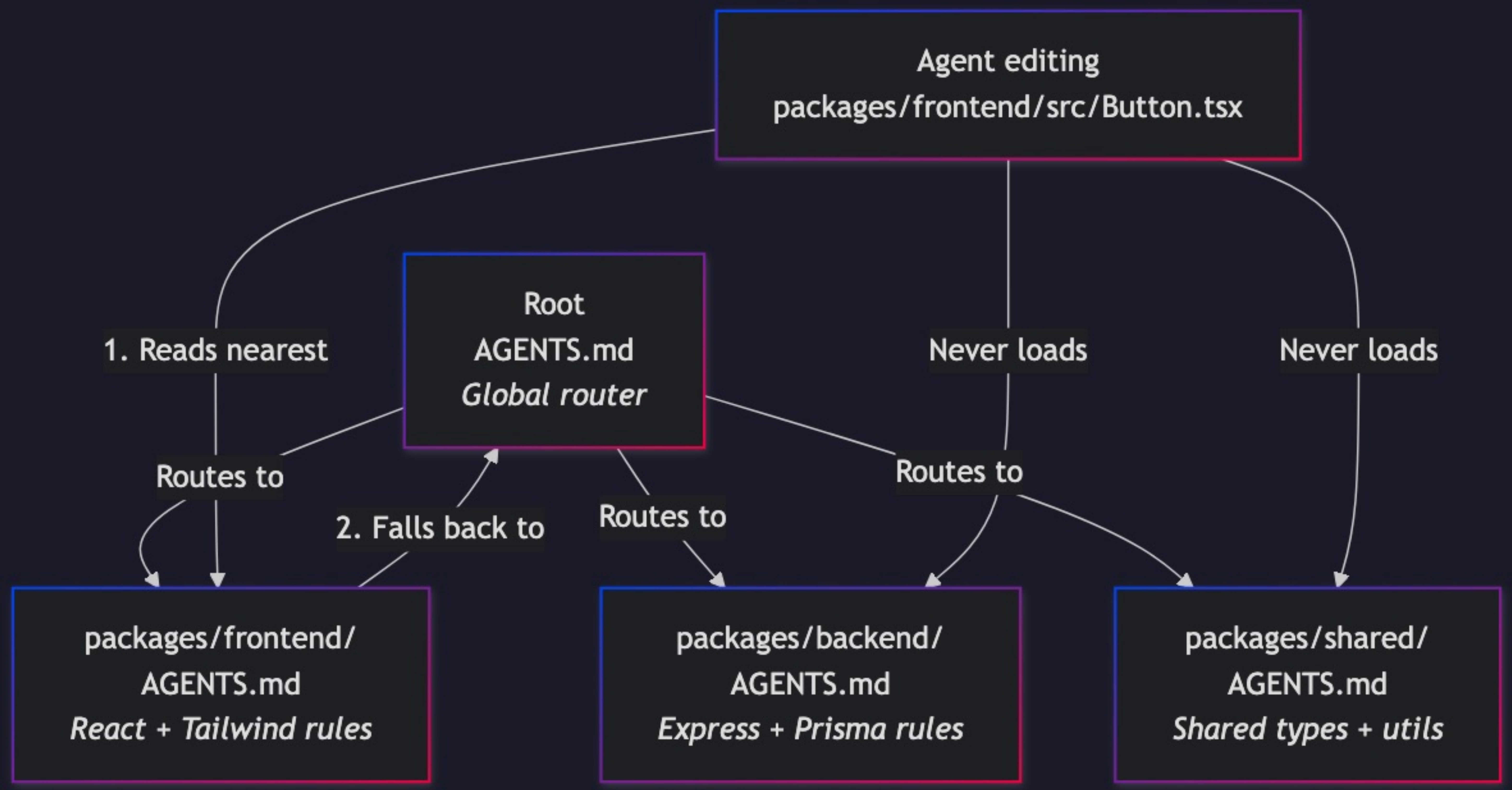
IDE plugin (sub-second, surgical) vs. autonomous cloud agent (full VM, large-scale). Complementary, not competing.



APPENDIX E

Nearest-File Precedence

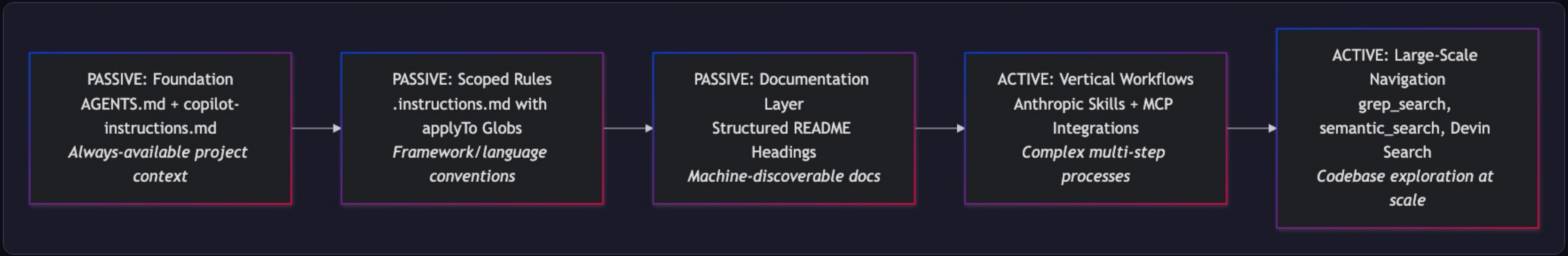
The root AGENTS.md routes. Each package overrides. Zero context leakage between domains.



APPENDIX F

The Hybrid Strategy Stack

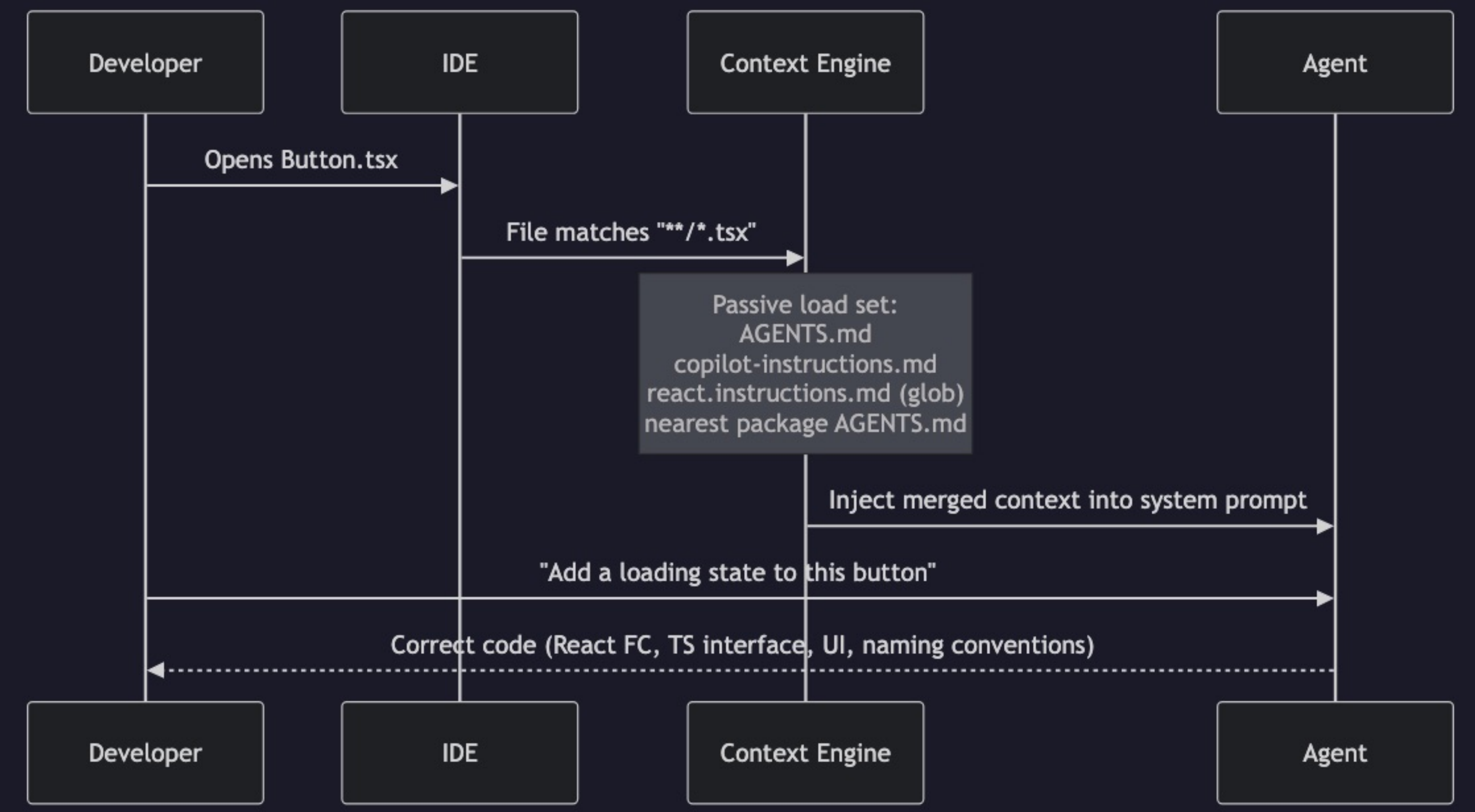
Five layers from foundation (passive, always-present) to large-scale active navigation. Build from the bottom up.



APPENDIX G

The Passive Steering Workflow

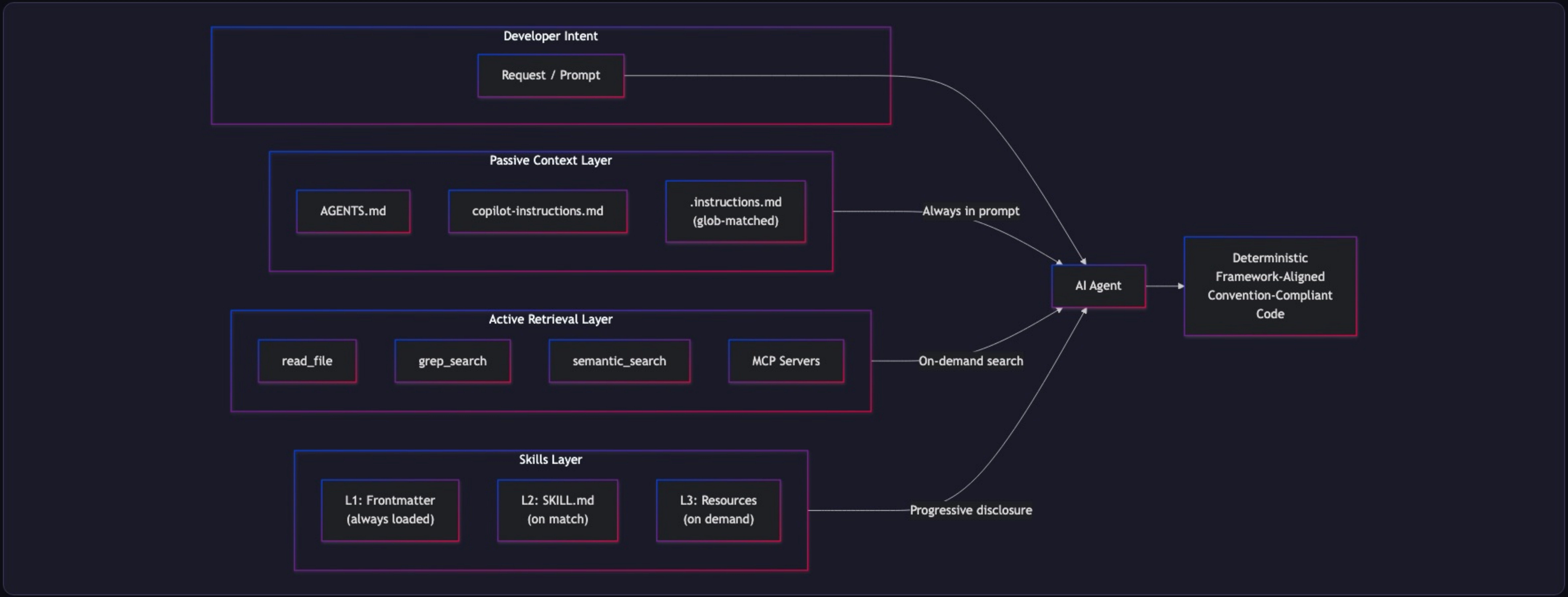
From file open → glob match → context merge → deterministic output.



APPENDIX H

The Complete Context Architecture

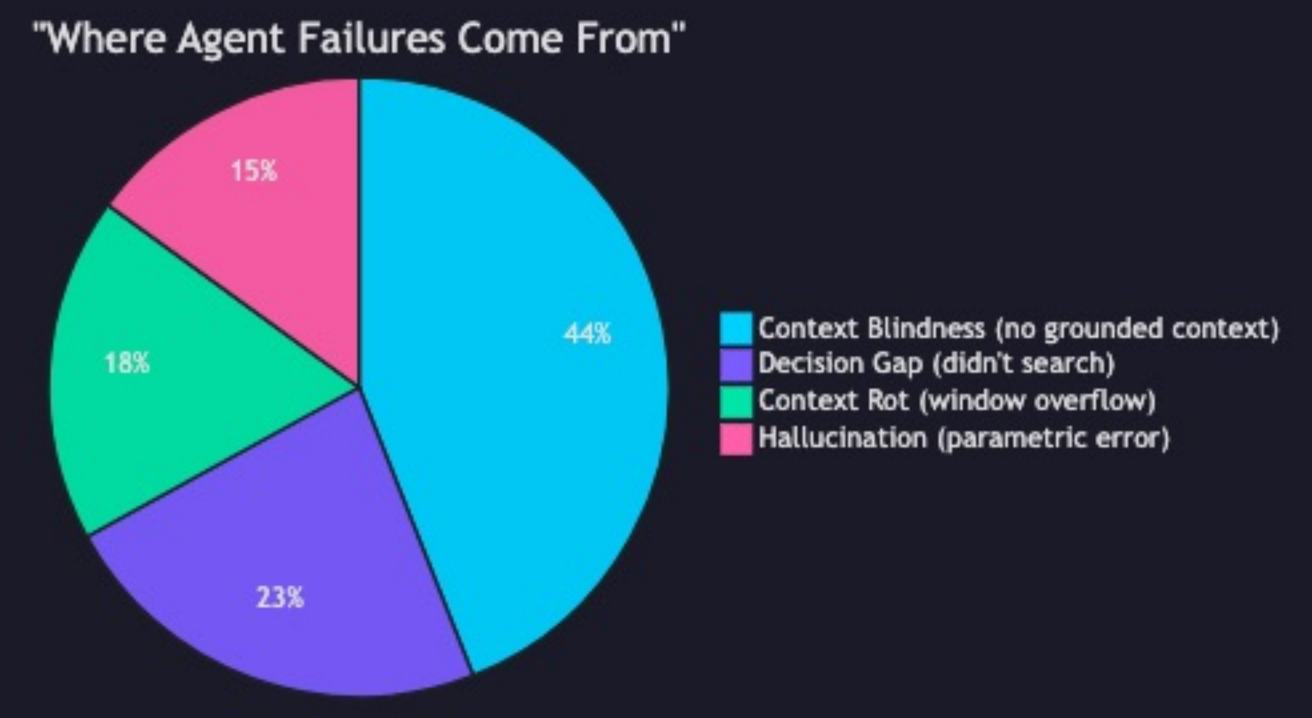
Three layers → one agent → deterministic, convention-compliant code.



APPENDIX I

Key Metrics Summary

Where agent failures originate — context blindness dominates at 44%.



44%
Context Blindness — no grounded context provided

23%
Decision Gap — agent didn't search for docs

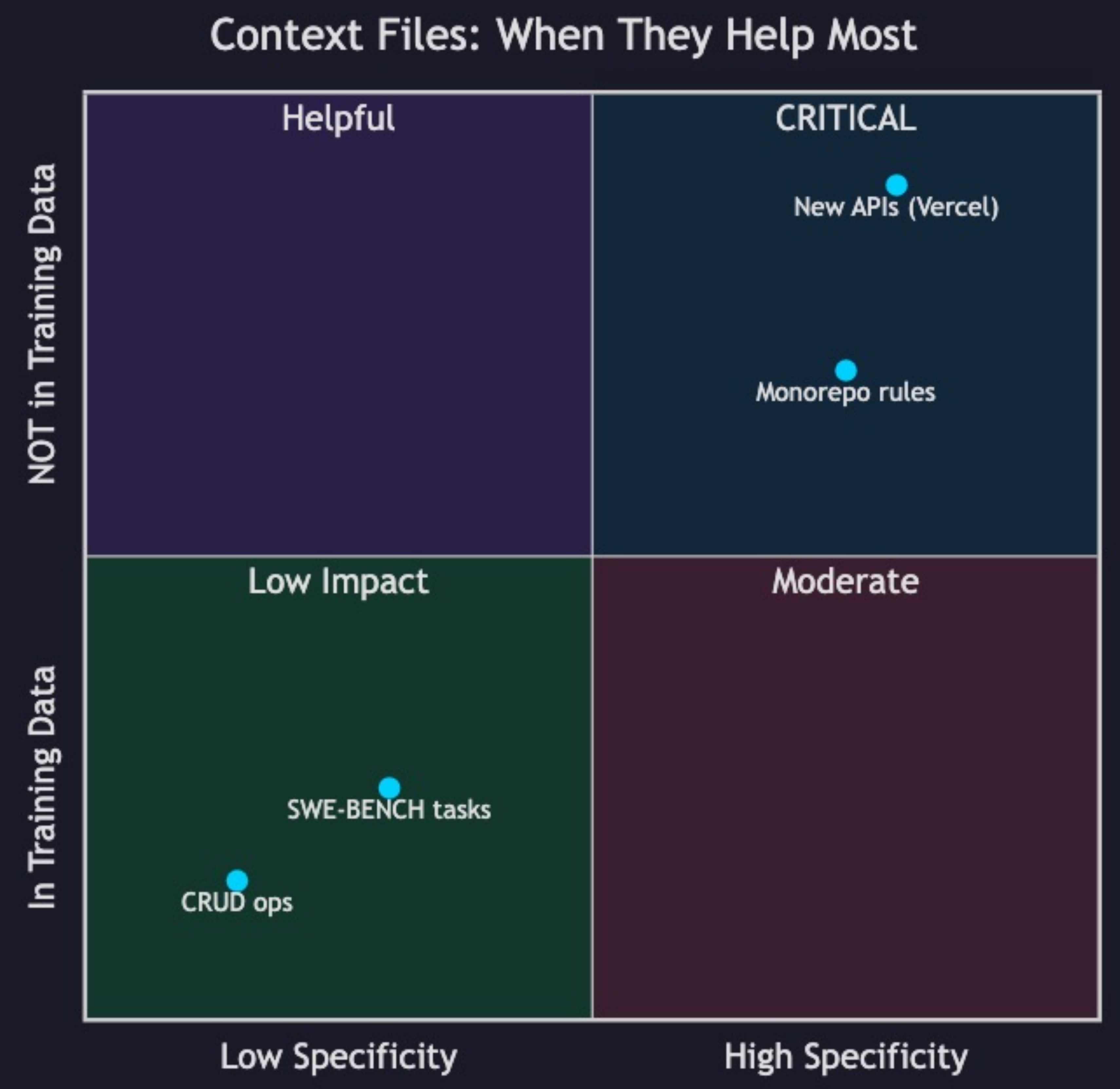
18%
Context Rot — window overflow discards rules

15%
Hallucination — parametric knowledge error

APPENDIX J

Context File Impact by Scenario

When context files help most: high project specificity + knowledge absent from training data.



INTERPRETATION

Top-right quadrant = **maximum AGENTS.md impact**. Bottom-left = minimal value. This explains the Vercel vs ETH Zürich discrepancy.

APPENDIX K

Full Presentation Timeline

15 minutes across 3 sections: Practical, The Why, Close — plus Q&A.